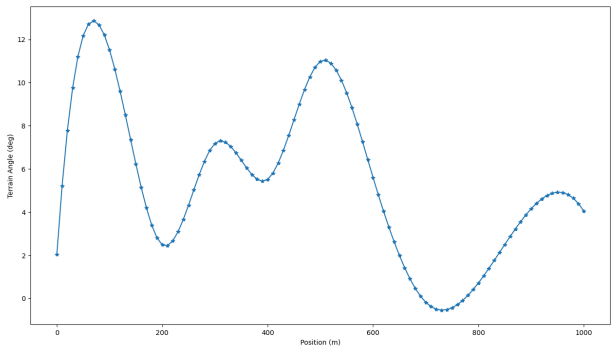
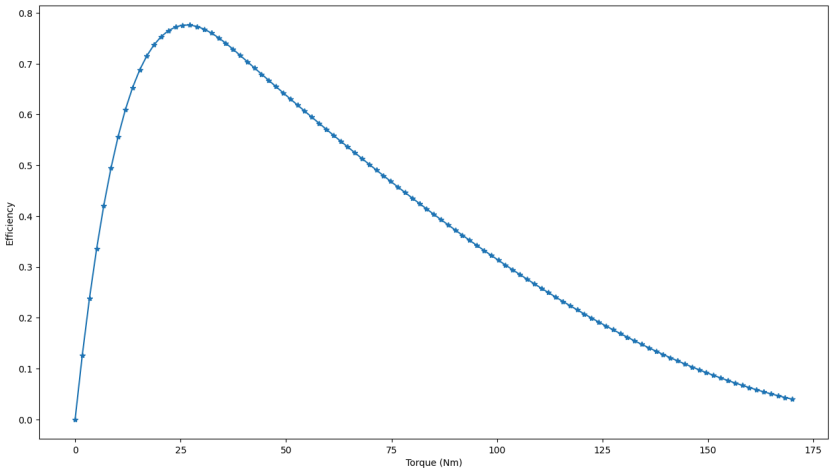
MEEN357 Phase 2 Report

Visualizing the Terrain



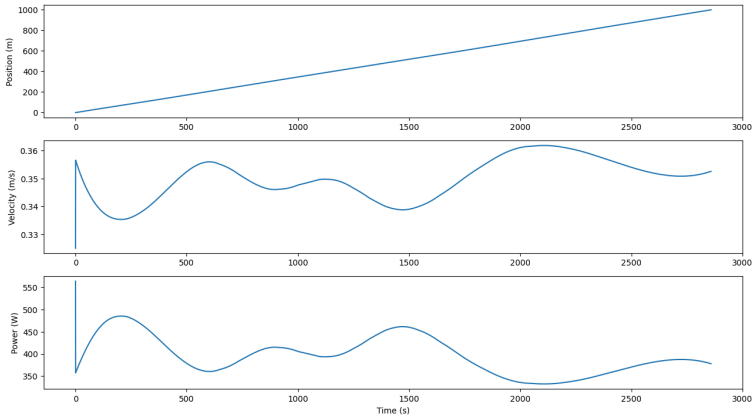
The graph to the right visualizes the terrain angle with respect to position. It should be noted that the graph does not represent the look of the terrain itself. This curve is smooth and goes through every point due to the usage of a cubic spline interpolation, forcing first and second derivatives to be equal at each point ensuring smoothness.

Visualizing Motor Efficiency



The efficiency of the motor never crosses the 78% mark at the top of the efficiency curve at 25Nm. This efficiency curve steeply increases as the motor is applying low amounts of torque then decreases as it applies more torque. Our curve is smooth because of the interpolation used to create data points between the torque-efficiency data points already given. This curve is similar to the curve seen in Figure 2 before the peak efficiency, but it differs from Figure 2 by being concave up rather than concave down after the peak efficiency.

Rover Simulation



The graphs here show the position, velocity, and power consumption (taking into account motor efficiency) of the rover with respect to time. The position graph can be assumed to be linear, since the variation in velocity is so small, but in reality it is very slightly nonlinear. Looking at the velocity and power plots, a decrease in velocity typically corresponds to an increase in power consumption. This is because the power required to move the rover is directly proportional to torque produced by the motors, which has an inverse relationship with rotational velocity and translational velocity. The velocity is related to the experimental terrain in that a higher terrain angle corresponds to a lower velocity. This velocity graph is smooth (aside from the starting zero velocity initial condition) due to the usage of a cubic spline interpolation on the terrain angle. The table below summarizes additional telemetry data from this simulation.

Completion Time (s)	2858.571464286607
Distance Traveled (m)	999.8388331389903
Maximum Velocity (m/s)	0.361807947514746
Average Velocity (m/s)	0.34972997508642323
Battery Energy Consumed (J)	1132512.197912361
Energy per Distance (J/m)	1132.6947507698249

Analysis of Energy Needs

Experiment1 ends on the condition of distance reaching 1000m while consuming 1132512.2 Joules from the battery, which is greater than the 0.9072e6 Joules that the Lipo battery can store. Therefore, the rover cannot complete the test case defined in experiment1.